

Machine learning driven exploration of hydride superconductors at ambient pressure

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ABSTRACT

Hydrogen-rich materials are among the most promising candidates for achieving high-temperature superconductivity due to their light atomic mass and strong phonon-mediated Cooper pairing. While many high-temperature superconducting hydrides have been reported experimentally, their practical applicability remains limited due to the required extreme conditions, motivating the search for stable or metastable superconducting phases at ambient pressure. Here, we combine *ab initio* electron-phonon coupling calculations with machine learning methods to explore over two million hydride structures, identifying more than 600 compounds with predicted critical temperatures (T_c) above 20 K. This dataset reveals a large chemical and structural diversity among potential superconductors, including perovskites, kagome lattices, and compounds with **isolated hydrogen octahedra**, among others. Despite this diversity, high- T_c materials consistently exhibit substantial hydrogen contributions to the electronic density of states at the Fermi level. **Symbolic regression analysis quantitatively confirms this correlation**. Thermodynamic analysis shows that all identified superconductors lie above the convex hull of stability, with energies typically exceeding 100 meV/atom above the hull. This implies that experimental synthesis can only be obtained through non-equilibrium approaches, including high-pressure or high-temperature methods followed by quenching, or by thin-film deposition techniques. Moreover, as many promising candidates resemble degenerate semiconductors, a possible synthesis route may involve controlled doping strategies of parent semiconducting compounds. This work substantially expands the known landscape of hydride superconductors and establishes design principles to guide future experimental realization.

1. Introduction

Superconductivity is a quantum phenomenon where materials exhibit exactly zero electrical resistance and expel magnetic fields below a critical temperature (T_c). Since its discovery in 1911 [1], it has captivated scientific interest, as it opens the door to a wide range

of technological applications. The quest for high-temperature superconductivity has been one of the most challenging pursuits in condensed matter physics for decades. Among various materials, hydrogen-rich compounds have emerged as particularly promising candidate superconductors.

The theoretical foundation for exploring hydride superconductors was established by Ashcroft [2], when he proposed that metallic

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hydrogen could exhibit high-temperature superconductivity due to its light mass and high phonon frequencies. Recognizing that the extreme pressures required for metallic hydrogen were experimentally challenging, Ashcroft later introduced the concept of “chemical pre-compression” [3]. This ingenious idea suggested that hydrogen-rich chemical compounds could effectively compress hydrogen through chemical bonding with heavier elements, achieving the beneficial electronic properties of dense hydrogen at substantially lower pressures. This redirected experimental efforts toward synthesizing metal hydrides under high-pressure conditions, ultimately leading to the discovery of several high-temperature hydride superconductors [4].

In 2015, sulfur hydride (H_3S) was experimentally found to exhibit a T_c of 203 K at around 150 GPa [5], being confirmed by the observation of zero resistance, a substantial isotope effect, and a magnetic field dependence of T_c . Another milestone appeared in 2019, marked by the observation of superconductivity LaH_{10} near 250 K at 170 GPa [6,7]. Each of these experimental breakthroughs has, without exception, been motivated by prior *ab initio* studies [8–11]. The success of *ab initio* predictions in guiding experimental discoveries of high- T_c superconducting hydrides has stimulated a large-scale search for hydrogen-based superconductors, many of which have been experimentally validated such as YH_6 [12,13], YH_9 [13] and CaH_6 [14,15] with T_c exceeding 200 K. In particular, room temperature superconductivity with a high T_c of 298 K has been experimentally reported in $\text{LaSc}_2\text{H}_{24}$ [16], validating the prior theoretical predictions from the same laboratory [17]. In spite of the high T_c of these synthesized hydrogen-rich compounds, they all require extremely high pressures, often exceeding 150 GPa to stabilize their high- T_c superconducting phases. This presents a significant obstacle for practical applications, where materials are expected to superconduct at ambient or near ambient pressure.

In contrast to the complex nature of the unconventional superconductors such as the copper-based cuprates [18], iron-based pnictides [19], and nickelates [20,21] where an accurate predictive theory is lacking, the superconducting hydrides are expected to be conventional superconductors following the theory developed by Bardeen, Cooper, and Schrieffer (BCS) [22] and its extensions formulated by Migdal and Eliashberg [23]. Favorable conditions for the classical mechanism include the fact that the light mass of hydrogen yields high phonon frequencies that enhance T_c , as well as the strong covalent H–H and metal–H bonding that generates a large electron–phonon coupling constant λ . Moreover, the predominantly s/p character of electronic states near the Fermi level disfavors strong correlation effects that could drive unconventional pairing. Therefore, there are well-established computational methods, in conjunction with suitable approximations, that can predict and explain superconductivity in the hydrogen-based compounds. In practice, however, the accurate prediction of superconducting properties from first principles remains computationally demanding. *Ab initio* approaches based on density functional theory (DFT) and the Migdal–Eliashberg formalism require the evaluation of electron–phonon coupling constants and phonon spectra across dense grids in reciprocal space. This renders systematic high-throughput studies of conventional superconductivity for large material families or complex structural configurations prohibitively expensive.

In this context, recent advances in high-throughput and machine learning (ML) techniques have opened new avenues for accelerating the discovery of superconducting materials. By leveraging large datasets, ML models can rapidly screen vast chemical spaces and predict key properties without resorting to full *ab initio* calculations at every step [24,25]. Recently, some of us used a ML-accelerated framework to screen more than 150 thousand compounds for conventional superconductivity at ambient pressure [26], identifying more than 50 compounds with predicted superconducting transition temperatures above 20 K.

In the present study, we **take a leap forward**, and extend these investigations by conducting a large-scale search for high- T_c superconductivity in over two million candidate hydride structures. This ML

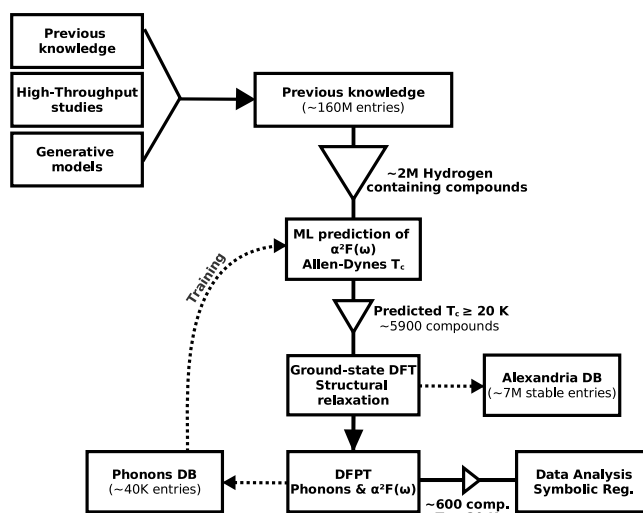


Fig. 1. Schematic overview of the complete workflow employed in this work, from the initial data sources to candidate filtering and final analysis. The initial dataset is constructed from previously reported studies and experimental data, complemented by structures generated using our in-house generative model [29], yielding an initial pool of approximately two million hydride compounds. These structures are screened using an ALIGNN-based model to predict the Eliashberg spectral function and to estimate the superconducting critical temperature via the Allen–Dynes formula, resulting in about 5900 candidates with a predicted $T_c \geq 20$ K. These candidates are subsequently subjected to ground-state DFT structural relaxation and DFPT calculations of phonon spectra and electron–phonon coupling. The final analysis is performed on the subset of compounds with a calculated $T_c \geq 20$ K.

accelerated approach resulted in the identification of more than 600 compounds with T_c above 20 K, exhibiting a broad range of chemical compositions and crystal structures, and displaying diverse bonding environments and structural motifs. In the following, we provide a detailed analysis of these candidates, examining their predicted superconducting properties, structural characteristics, and the underlying chemical and physical trends, thereby offering insights to guide future experimental and theoretical studies.

2. Methods

We employed a ML accelerated workflow (see Fig. 1) analogous to that introduced in Ref. [26], designed to efficiently screen large chemical spaces for promising superconducting hydrides. Our starting point is a dataset of approximately 2 million hydrogen-containing materials from either the Alexandria Database [26–28] or suggested by MatraGenoa [29], our in-house generative machine learning model [29]. In the latter case, we asked the model to generate compounds that were relatively close to the convex hull across the full chemical and structural range, and we did not condition the model any further. These generated structures were then relaxed with ORB-v2 [30]. For interested readers seeking details of these machine learning models, we refer to the original publications [29–31]. The superconducting transition temperature of these relaxed compounds was then estimated using another ML model, based on ALIGNN [32].

Critically, the input to the ALIGNN model consists solely of the chemical composition and the crystal structure. ALIGNN does not require hand-crafted electronic or phononic descriptors as input. Instead, it automatically learns relevant structural features through the graph neural network architecture. The node features for each atom include intrinsic atomic properties, for example, atomic number, electronegativity, atomic radius, number of valence electrons, first ionization energy, and electron affinity. These are static properties from the periodic table and do not require electronic structure calculations. The edge

features encode interatomic distances for all atom pairs within a cutoff radius (8 Å in our model) up to a maximum of 12 neighbors per atom. The line graph then creates connections between edges (bonds) that share a common atom, allowing the model to learn about bond angles and local coordination environments. **Rather than directly predicting T_c , our ALIGNN model predicts an eight-dimensional latent representation of the full Eliashberg spectral function $\alpha^2 F(\omega)$. This latent vector is then decoded using a trained autoencoder to reconstruct the spectral function, from which we compute λ , $\omega_{\log}$, and T_c via standard formulas. This two-stage approach proved substantially more accurate than direct T_c prediction, reducing the MAE from 1.3 K to 1.1 K in validation tests. The physical insight is that $\alpha^2 F(\omega)$ encodes richer information about the electron–phonon interaction than the scalar quantities λ and $\omega_{\log}$, and this intermediate representation helps the model generalize better.**

The loss function used was the root-mean-square error (RMSE) between predicted and reference latent vectors. To decode the latent representation back into the full spectral function, we employed a symmetric autoencoder trained on the same dataset. **The autoencoder loss function combined the RMSE of the reconstructed $\alpha^2 F(\omega)$ and λ , with equal weighting.** Although the training loss did not explicitly include the superconducting critical temperature, the hyperparameters of the autoencoder were tuned to minimize the prediction error in T_c . This indirect optimization yielded improved performance compared to a direct prediction of T_c using ALIGNN alone.

In the training set we included not only intermetallic compounds, but also hydrides in order to improve the predictive power for this class of materials. In total, the model was trained on a dataset of 9800 compounds. This set was partitioned using an 8:1:1 ratio for training, validation, and testing, respectively. The model achieved mean absolute errors of 0.08 for the electron–phonon coupling constant λ , 23 K for the logarithmic average phonon frequency $\omega_{\log}$, and 1.1 K for the superconducting transition temperature T_c . These values compared favorably to the mean values of the target distributions (0.49, 211 K, and 3.1 K, respectively) [33].

All compounds with a predicted superconducting critical temperature $T_c \geq 20$ K were then selected for electron–phonon coupling calculations using density-functional perturbation theory (DFPT).

From $\alpha^2 F(\omega)$, the electron–phonon coupling constant and logarithmic average phonon frequency were computed as:

$$\lambda = 2 \int_0^\infty \frac{\alpha^2 F(\omega)}{\omega} d\omega, \quad (1)$$

$$\omega_{\log} = \exp \left[\frac{2}{\lambda} \int_0^\infty \frac{\alpha^2 F(\omega)}{\omega} \ln(\omega) d\omega \right], \quad (2)$$

$$\omega_2 = \left[\frac{2}{\lambda} \int_0^\infty \omega \alpha^2 F(\omega) d\omega \right]^{\frac{1}{2}}. \quad (3)$$

The superconducting transition temperature was estimated using the Allen–Dynes modified McMillan equation [34]:

$$T_c = f_1 f_2 \frac{\omega_{\log}}{1.2} \exp \left[-\frac{1.04(1 + \lambda)}{\lambda - \mu^*(1 + 0.62 \lambda)} \right], \quad (4)$$

where the strong-coupling and shape correction factors are defined as:

$$f_1 = \left(1 + \left[\frac{\lambda}{2.46(1 + 3.8\mu^*)} \right]^{2/3} \right)^{1/3}, \quad (5)$$

$$f_2 = 1 + \frac{\lambda^2(\omega_2/\omega_{\log} - 1)}{\lambda^2 + [1.82(1 + 6.3\mu^*)(\omega_2/\omega_{\log})]^2}. \quad (6)$$

Throughout this work, the Coulomb pseudopotential was fixed at $\mu^* = 0.10$. This value fits well within the range of values used in literature [35,36]. In total we performed approximately 5900 DFPT calculations, from which we identified more than 600 dynamically stable materials with $T_c \geq 20$ K. By comparing ALIGNN predictions to DFPT-calculated values for the 600 validated hydrides with $T_c \geq 20$ K, we obtain MAE values of 0.21 for λ , 70.4 K for $\omega_{\log}$, and 6.5 K for T_c .

While these errors are larger than the test set performance (reflecting the extrapolation to the high- T_c regime), they remain acceptable for a screening tool. The error in T_c (6.5 K) is smaller than our selection threshold of 20 K, confirming that the model successfully enriches the candidate pool with genuine high- T_c superconductors.

3. Results

In Table 1, we list all compounds that emerged from our high-throughput search with $T_c \geq 50$ K. Some of the remaining compounds with $T_c \geq 20$ K will be discussed below, and a complete list, including the crystal structure, distance to the convex hull of thermodynamic stability, electron and phonon band structures, Eliashberg spectral function $\alpha^2 F(\omega)$, etc. can be found in the Supplementary Information (SI). In Table 1, we also present the space group number, the distance to the convex hull of stability (E_{hull}), the logarithmic average of the phonon frequencies ($\omega_{\log}$), the electron–phonon coupling constant (λ), the transition temperature (T_c), and an identification number that can be used to easily find the compound in the SI. Naturally, some of these compounds have been previously reported in the literature. In such cases, we also show the reference to the original work in the table.

3.1. Statistical analysis of superconducting hydrides

We begin by analyzing our results from a global perspective, focusing on the overall trends of the complete dataset rather than on individual compounds. This approach enables the identification of general correlations between structural, chemical, and superconducting properties across the diverse set of hydride candidates, providing a foundation for the subsequent, more detailed examination of specific systems.

Fig. 2 shows a heatmap of the elemental distribution across all hydrides exhibiting $T_c > 20$ K. The distribution highlights the chemical diversity of the identified superconductors, and includes elements from multiple regions of the periodic table, displaying the wide compositional space in which conventional superconductivity can emerge under suitable structural and electronic conditions. Among metallic elements, the noble metals Cu, Pd, Au, and Pt are strongly represented among the superconducting compounds. To this we can add, from group IIB, Zn and Cd, and, from group IIIA, Al, Ga, In, and Tl. As we will see in the following, these are the chemical elements that usually bind directly to hydrogen. We also find a few lanthanide-containing compounds, indicating that high- T_c at ambient pressures is also possible in systems with f -electrons. On the contrary, very few compounds are found with early transition metals. Alkaline and alkaline-earth metals also occur frequently, often serving as electron donors that stabilize hydrogen-rich frameworks. The occurrence decreases in general with the period of the periodic table, indicating a greater difficulty to accommodate larger ions in the superconducting structures. As expected, we find very few systems with other non-metals, although a hand-full of mixed anion compounds can be found in the dataset.

In our dataset, more than 600 hydrides exhibit predicted critical temperatures above 20 K. The distribution of their stability and superconducting properties is shown in Fig. 3. To further explore the relationship between the electron localization function (ELF) [44–46] networking parameter ϕ and T_c , discussed by Belli et al. [47], we also include the distribution of ϕ .

Overall, E_{hull} is relatively high, as expected for metallic hydrides, following an approximately **normal distribution** centered around 200 meV/atom, with a right-hand tail extending up to about 1300 meV/atom ($\text{B}_6\text{BiNi}_6\text{H}$, SI #625). In contrast, only a few hydrides show low energies above the hull, and none lie on the convex hull. The most thermodynamically stable compound in the dataset is the experimentally known PdH (SI #611) [48,49], with an energy of 35 meV/atom above the hull.

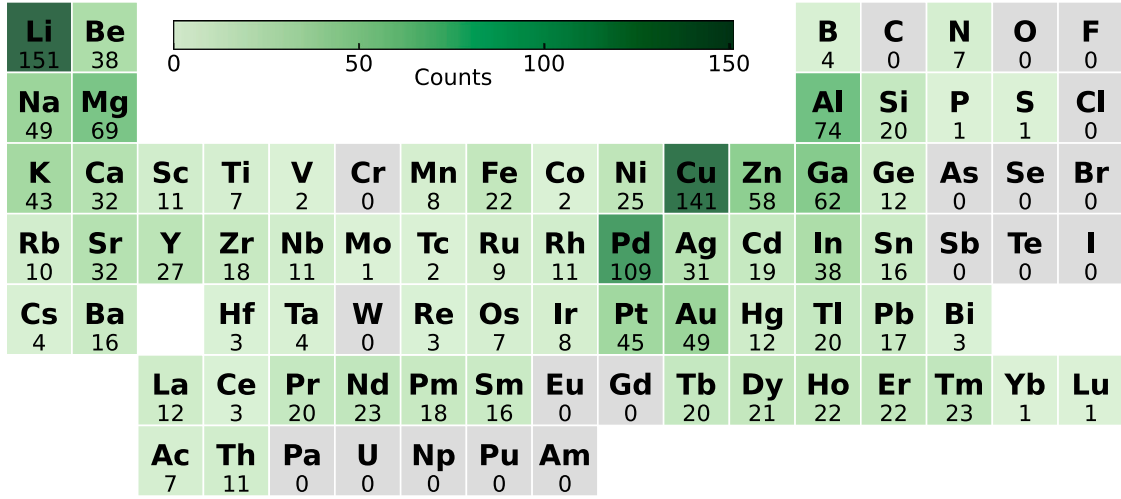


Fig. 2. Periodic table heatmap for chemical element occurrence in the hydrides present in our dataset. Non-represented elements are depicted in gray.

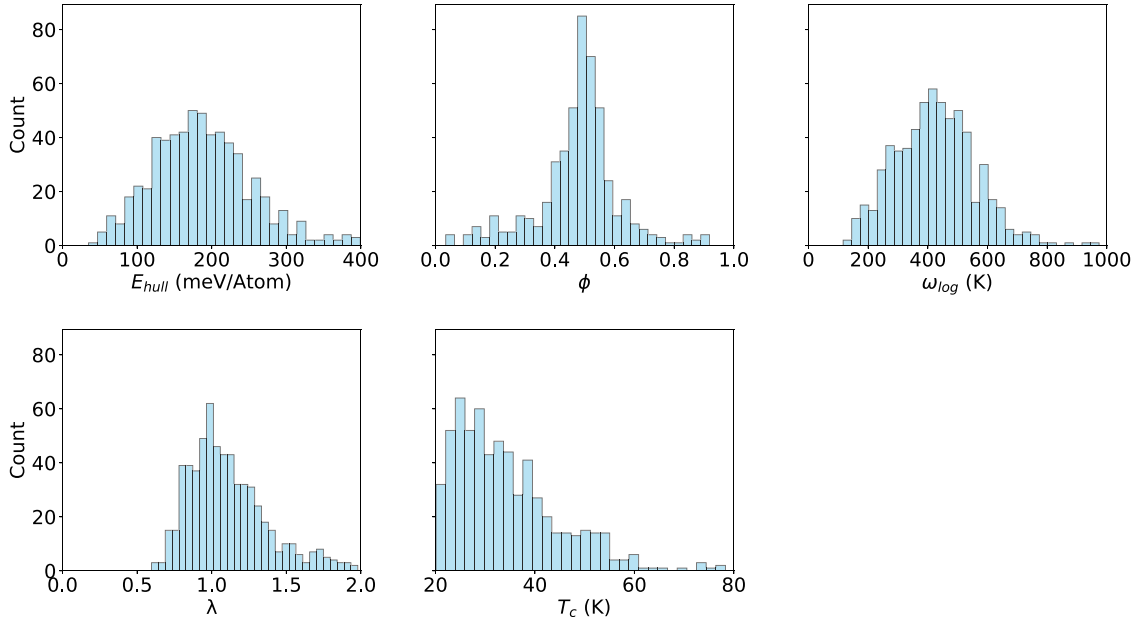


Fig. 3. Distribution of thermodynamic, electronic and superconducting properties across the dataset. Each row shows histograms for (from left to right) E_{hull} (in meV/atom), ϕ , ω_{log} (in K), λ and T_c (in K).

We defined ϕ as the ELF networking value, corresponding to the highest value of the ELF that creates an isosurface spanning through the whole crystal in all three Cartesian directions [47]. The distribution of these values is quite symmetric and centered at 0.5. Belli et al. [47] demonstrated a tendency toward higher critical temperatures with increased ELF values. However, our workflow consistently yielded ELF values around 0.5, with few datapoints having ELF network parameters ϕ outside the range $0.4 < \phi < 0.6$.

Both ω_{log} and λ have distributions centered at relatively high values (as expected for high- T_c superconductors), each with a long tail toward larger magnitudes. The coupling constant λ reaches up to 3.1 (OsAl₂H₇, SI #171), while ω_{log} has a maximum of 974 K for Na₂BePdH₆ (SI #282). Although these correspond to high predicted transition temperatures, some hydrides exhibit relatively low λ values. For instance, Na₂BePdH₆ (SI #282) has the smallest λ (0.6) yet the largest ω_{log} . Conversely, the lowest ω_{log} value of 116 K found for Ta₄H₁₅ (SI #111) is compensated by a very high λ of 2.7.

The critical temperature distribution resembles a chi-squared-like curve, with a rapidly decaying tail at higher T_c values, reaching up

to 112 K (PtH₄, SI #559) [37]. The drop near 20 K on the left side likely results from our workflow: we first compute T_c with lower accuracy, and only materials with $T_c \geq 20$ K undergo high-accuracy recalculations, thus excluding lower values from the final dataset.

Fig. 4 shows a heatmap of the Pearson correlations between E_{hull} , ω_{log} , λ , and T_c with various structural and electronic features of our dataset. Overall, the correlations are weak, reflecting both the complexity of high- T_c superconductivity in hydrides and the inherently non-linear relationships among these quantities. Thermodynamic stability shows positive correlation with ELF parameters and weaker correlation with hydrogen content and crystal density, though these trends remain difficult to interpret physically. The influence of hydrogen on the superconducting behavior is evident in the correlations involving the fraction of the hydrogen-projected DOS. While ω_{log} shows a strong positive correlation with these hydrogen-related quantities, λ exhibits almost no dependence. This distinction arises because ω_{log} primarily reflects the characteristic phonon frequency, which increases with greater hydrogen participation due to its low atomic mass. In contrast, λ decreases for higher-frequency modes and is instead governed

Table 1

All hydrides found in this work with critical temperatures T_c (in K) above 50 K. Also shown are the space group number (spg), the distance to the convex hull of stability E_{hull} (in meV/atom), the logarithmic average phonon frequency ω_{log} (in K), the electron–phonon coupling constant λ , the transition temperature T_c , and an number used to identify the compound in the SI. Compounds with an energy above the hull under 100 meV/atom are highlighted in bold. The formulas use the IUPAC convention that follows loosely the electronegativity scale. The transition temperatures are calculated with the Allen-Dynes formula, including the correction factors [34]. In the cases where the structure has been proposed previously in the literature, we also indicate the corresponding reference.

Formula	spg	E_{hull}	ω_{log}	λ	T_c	Ref.	#ID
PtH ₄	229	147	733	1.74	112.2	[37]	SI #559
PdH ₄	229	319	570	2.11	107.0	[37]	SI #560
Li ₃ AuH ₆	225	172	567	1.75	88.4	[38]	SI #199
Li ₂ CuH ₆	225	204	490	1.96	86.0	[39]	SI #198
Mg ₂ PtH ₆	225	192	770	1.24	78.3	[40]	SI #191
SrBe ₄ H ₁₂	229	333	574	1.54	76.7		SI #34
PtPdH ₈	221	227	573	1.51	76.1		SI #169
Li ₃ BeH ₆	167	214	618	1.40	74.2		SI #226
NaLi ₂ AuH ₆	225	150	485	1.72	74.0		SI #277
KInH ₃	221	115	487	1.70	72.9	[26]	SI #587
Al ₂ RuH ₅	107	190	436	1.78	69.4		SI #539
Be ₃ H	164	203	588	1.34	65.9	[41]	SI #619
Mg ₂ PdH ₆	225	196	760	1.08	63.8	[40]	SI #187
MgCu ₃ S ₂ H ₁₂	166	436	388	1.78	61.8		SI #26
Li ₂ LaAuH ₈	164	159	546	1.31	60.8		SI #147
Li ₃ ZnH ₆	167	196	260	2.61	60.8		SI #223
K ₂ AgTiH ₆	225	148	396	1.71	60.0		SI #270
AlAgH ₆	166	320	396	1.69	59.5		SI #380
LiTmPdH ₆	187	199	507	1.37	59.5		SI #462
Mg₂IrH₆	225	85	634	1.16	59.4	[42]	SI #185
Li ₂ CaAuH ₈	164	194	522	1.31	58.3		SI #145
Li ₂ YCuH ₈	164	168	552	1.25	57.9		SI #132
Na ₂ CuHgH ₆	225	149	574	1.21	57.3		SI #272
Al ₂ OsH ₇	225	265	204	3.17	57.3		SI #171
LiErPdH ₆	187	191	502	1.30	55.7		SI #460
Li ₂ AcAuH ₈	164	146	583	1.17	55.6		SI #150
Rb ₂ AgZnH ₆	225	194	354	1.77	55.3		SI #257
Al ₂ OsH ₅	107	149	439	1.44	55.1		SI #540
Rb ₂ AgZn ₃ H ₁₂	166	201	377	1.64	54.9		SI #23
Mg ₂ PtH ₅	107	101	675	1.05	54.8		SI #537
LaCu ₃ H ₁₀	225	285	420	1.49	54.8		SI #58
Li ₂ DyCuH ₈	164	169	523	1.25	54.6		SI #139
Li ₂ TbCuH ₈	164	163	541	1.22	54.6		SI #138
AgH ₃	221	422	562	1.18	54.3		SI #581
Li ₂ HoCuH ₈	164	178	497	1.30	54.8		SI #140
Li ₂ ErCuH ₈	164	180	469	1.35	54.2		SI #141
MgCuSnH ₆	149	229	507	1.27	54.1		SI #523
Y ₃ ZrH ₁₅	5	135	581	1.15	54.0	[43]	SI #14
Mg₂RhH₆	225	50	771	0.96	53.8	[40]	SI #188
CaAuH ₆	166	205	380	1.60	53.7		SI #384
Rb ₂ CdTiH ₆	225	154	427	1.45	53.4		SI #275
Y ₄ H ₁₅	220	233	474	1.32	53.1	[43]	SI #4
Li ₂ TmCuH ₈	164	186	430	1.42	52.9		SI #142
NSn ₂ H ₇	38	316	324	1.82	52.8		SI #174
LiCaCuH ₆	187	205	450	1.36	52.5		SI #411
Li ₂ SmCuH ₈	164	144	584	1.12	52.4		SI #137
Na ₂ AuHgH ₆	225	143	498	1.25	52.4		SI #273
LiHoPdH ₆	187	185	497	1.25	52.0		SI #458
NaInSnH ₆	149	262	366	1.60	51.9		SI #522
Tb ₂ TiH ₈	71	248	367	1.60	51.8		SI #86
KCuCdH ₆	187	278	587	1.11	51.7		SI #413
Li ₂ SrAuH ₈	164	145	627	1.06	51.7		SI #146
Na ₂ BeH ₆	225	376	299	1.92	51.6		SI #201
Lu ₄ H ₁₅	220	225	523	1.19	51.3	[43]	SI #12
Li ₂ PmCuH ₈	164	139	594	1.09	51.3		SI #136
TmCu ₂ H ₆	65	218	462	1.31	51.3		SI #219
Yb ₄ H ₁₅	220	292	499	1.23	51.1	[43]	SI #13
Na ₃ AgZn ₂ H ₉	164	124	624	1.06	50.8		SI #70
ErCu ₂ H ₆	65	208	476	1.27	50.7		SI #218
Mg₂NiH₅	107	66	542	1.15	50.5		SI #536
Tm ₄ H ₁₅	220	227	495	1.23	50.4	[43]	SI #9
Li ₂ PdSiH ₆	146	189	550	1.14	50.3		SI #363
Li ₂ NdCuH ₈	164	132	603	1.07	50.0		SI #135
KMg ₂ PdH ₈	164	117	560	1.12	50.0		SI #148

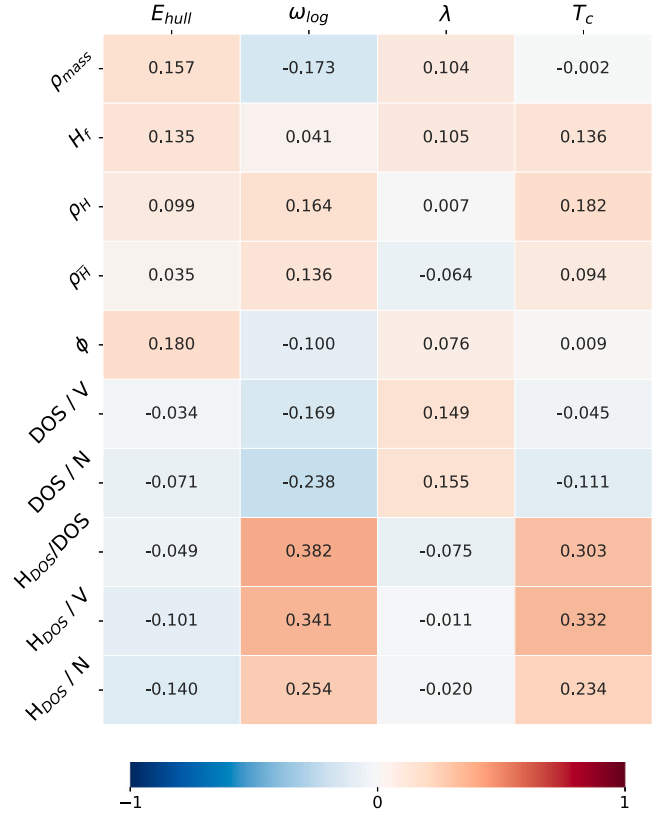


Fig. 4. Correlation heatmap between E_{hull} (meV/atom), ω_{log} (K), λ , and T_c (K) with various features: the mass density (ρ_{mass} in u.m.a./Å³), hydrogen content fraction (H_f), densities of hydrogen (ρ_H in atom/Å³) and non-hydrogen ($\rho_{\bar{H}}$ in atom/Å³), the ELF networking parameter (ϕ), the DOS at the Fermi level normalized by volume (DOS/V in state/eV/Å³) and number of atoms (DOS/N in state/eV/atom), and the projected hydrogen DOS similarly normalized to the total DOS ($H_{\text{DOS}}/\text{DOS}$), volume (H_{DOS}/V in state/eV/Å³), and number of atoms (H_{DOS}/N in state/eV/atom). Color scale indicates Pearson correlation coefficients.

by the electronic DOS near the Fermi level, which directly affects electron pairing. This behavior results in the stronger correlation observed between λ and the total DOS per unit volume. Furthermore, ω_{log} shows a negative correlation with the total DOS per volume. We note that we do not find any correlation between the value of the ELF parameter and ω_{log} and λ . These trends extend to the critical temperature, with the strongest T_c correlations corresponding to the factors most strongly influencing ω_{log} .

3.2. Symbolic regression

To further quantify the interplay between structural, electronic, and superconducting properties mentioned in the statistical analysis, we applied symbolic regression using the PySR package [50] to different sets of properties. This approach can unveil complex non-linear relationships that linear correlation analysis cannot capture. Three separate regression runs were performed, each targeting a different property: T_c , λ , and ω_{log} . The input features for all runs included the unit cell volume (V), unit cell mass (M), number of hydrogen atoms (N_H), number of non-hydrogen atoms ($N_{\bar{H}}$), DOS at the Fermi level, hydrogen-projected density of states at the Fermi level (H_{DOS}), and ELF networking parameter ϕ .

The best fit was considered the one with the larger derivative of the logarithmic of the loss with respect to complexity (i.e., the number of operations) and a root mean squared error (RMSE) below 1.5 times

the best RMSE obtained across all fits, following the default settings of PySR [50].

The best fits are as follows:

$$\omega_{\log} = 512 \times \left(\frac{H_{\text{DOS}}}{\text{DOS}} \right)^{0.165}, \quad (7a)$$

$$T_c = 369 \times \frac{H_{\text{DOS}}}{V - 7.59N_H} + 29.1 \quad (7b)$$

$$\lambda = 1.12 + \frac{0.00415}{\text{DOS} - 2.29} \quad (7c)$$

As expected, the symbolic regression successfully converged to expressions consistent with the observed correlations in Fig. 4. The model identified the dependence of $\omega_{\log}$ on hydrogen-related electronic descriptors, with the most consistent predictor being the hydrogen-projected density of states normalized by the total DOS. The resulting equation for T_c followed the same trend, showing a similar dependence on H_{DOS} normalized to the volume subtracted by the number of hydrogen atoms. These relations achieved absolute errors of 95 K for $\omega_{\log}$ (compared to 104 K for the mean of the dataset), 7.7 K for T_c (compared to 8.3 K for the mean), and 0.227 for λ (compared to 0.232 for the mean). These large errors relative to mean values reflect the fundamental difficulty of finding simple descriptors for superconductivity, even in well-understood electron-phonon systems.

We emphasize that the symbolic regression expressions are *not* intended to reproduce the Allen-Dynes formula; rather, they represent empirical approximations that relate λ , $\omega_{\log}$, and T_c directly to ground-state descriptors that are accessible from standard DFT calculations. The primary goal is to understand how superconducting properties depend on these descriptors and to capture the dominant physical trends governing high- T_c superconductivity in hydrides. The expressions given by Eqs. (7a)–(7c) should not be viewed as rigorous relationships.

3.3. Example superconducting hydrides

While there is no universal recipe for achieving high superconducting transition temperatures in hydrides, our screening revealed several families of compounds sharing similar chemical compositions or structural motifs, that consistently exhibit $T_c \geq 20$ K. For instance a substantial part of our dataset is composed of kagome materials, perovskites, both normal and inverted, double perovskites or vacancy ordered double perovskites, and structures that present isolated octahedral motifs. In the following, we provide an overview of representative compounds from these recurring families. For brevity, we exclude those systems that have already been thoroughly analyzed in previous works [26,40]. In the following “SM” denotes a simple metal, “TM” a transition metal or noble metal, and “LN” is a lanthanide.

3.3.1. Perovskites

Among the hydrides identified in our dataset, there are nine standard perovskites with the formula ABH_3 (see Fig. 5(a)) and two inverted perovskites where Mg and Al occupy the 3d Wyckoff position and hydrogen occupies the 1b site. Hydride perovskites have attracted considerable attention for hydrogen storage applications [51,52]. Recently, Du et al. predicted high- T_c superconductivity below 10 GPa for several systems based on the perovskite phase of KInH_3 , with 5 systems exhibiting T_c above 120 K [53]. Liu et al. studied hydrogen-metal bonded perovskites at ambient pressure and predicted T_c for several of these systems [54], while Li et al. [55] studied perovskites with structure MXH_3 , where M represents alkali and post-transition metals, and X denotes 3d, 4d, and 5d transition metals, identifying 2 systems with a T_c above 100 K. Several hydride perovskites have been synthesized experimentally [56], including AlSrH_3 [57] and $\text{SrPdH}_{2.7}$ [58]. Antiperovskite syntheses have likewise been reported, such as ChHM_3 (Ch = chalcogen; M = Li, Na) [59] and $\text{InH}_{0.8}\text{Pd}_3$ [60].

Within our dataset, we identify several hydride perovskites with promising superconducting properties. The compounds AlTiH_3

(SI #586) and AlInH_3 (SI #591) exhibit similar electronic and vibrational characteristics: their λ arises predominantly from hydrogen-related vibrational modes, and their Fermi levels are positioned near the shoulder of a DOS peak. In both materials, degenerate conduction bands form an electron pocket centered at the Γ -point, which transitions to a parabolic dispersion centered at the R point. These similarities are reflected in their comparable predicted critical temperatures of 35.4 K for AlTiH_3 and 38.1 K for AlInH_3 .

PbIrH_3 (SI #590) shares structural similarities with PbOsH_3 [40], differing only in the transition metal. However, it exhibits distinctly different phonon characteristics: the hydrogen-related vibrational modes are highly dispersive and extend into the low-energy region, particularly near the X point in the Brillouin zone, suggesting strong electron-phonon scattering in that region.

The inverted perovskite MgAl_3H (SI #621) displays a high electronic DOS at the Fermi level, predominantly from Al contributions. Its phonon spectrum contains well-separated manifolds, with high-frequency hydrogen modes positioned above the Al-dominated vibrations. These high-energy H modes contribute significantly to the electron-phonon coupling, yielding $\lambda = 1.42$ and $T_c = 44.4$ K.

3.3.2. Double perovskites

We identified 42 double perovskite compounds, with the typical formula $\text{A}_2\text{BB}'\text{H}_6$ (see Fig. 5(b)), in our dataset. This substantial number is consistent with the large number of simple perovskites also found in our screening. Hydride double perovskites have attracted attention for their potential in hydrogen storage [61] and as high-temperature ferromagnetic semiconductors [62]. Jiang et al. employed machine learning to screen nearly all possible hydride double perovskites, discovering 15 hypothetical systems predicted to be stable with T_c values up to 18.9 K [63]. Yao et al. [64] used an inverse material design generative framework to identify several hydride double perovskites, being 2 of those in our dataset: K_2LiCuH_6 (SI #237) and $\text{Na}_2\text{LiCuH}_6$ (SI #236). More recently, Wang et al. [65] screened several double perovskites with the structure A_2InZH_6 , where A represents an alkali metal and Z represents metals from groups IB to VA. The authors identified four high-temperature superconducting hydrides stable under ambient pressure, two of which are present in our dataset: K_2InAgH_6 (SI #241) and K_2InHgH_6 (SI #266). To the best of our knowledge, experimental synthesis of double perovskite hydrides at ambient pressure has not yet been achieved.

In the structures present in our data, the A sites are occupied by alkali metals, predominantly potassium and sodium, while the B and B' sites host typically metallic species. Due to the compositional diversity inherent in these quaternary systems, no consistent electronic or vibrational pattern emerges across the entire class.

The compound $\text{Na}_2\text{LiAuH}_6$ (SI #277) exhibits the highest predicted T_c within this family, reaching 74 K. Its electronic structure is relatively simple, with two bands intersecting the Fermi level, with a crossing at the Fermi energy between the K and Γ points, and two crossing bands overlapping between the L and Γ points. The DOS at the Fermi level is dominated by hydrogen states. The phonon spectrum exhibits low-frequency vibrational modes from Au and Na at the bottom, with these two manifolds well-separated from the higher-frequency Li and H modes. The high λ value of 1.7 arises primarily from the lower-energy hydrogen modes at 40 and 75 meV.

Interestingly, the $\text{Na}_2\text{LiCuH}_6$ chemical composition appears in our dataset in two different phases: a double perovskite with cubic symmetry (SI #236 [64]) and a trigonal phase with space group $P3m1$ (SI #357). Although their electronic structures are broadly similar, the trigonal phase displays a slightly reduced dispersion around the Fermi level, with a Dirac point at the K-point slightly above the Fermi level. The phonon spectrum of the perovskite phase is very similar to the one of $\text{Na}_2\text{LiAuH}_6$, with the same hydrogen modes contributing to the high λ value. With respect to the trigonal phase, the lithium modes are much more dispersed over the lower range of frequencies. The T_c

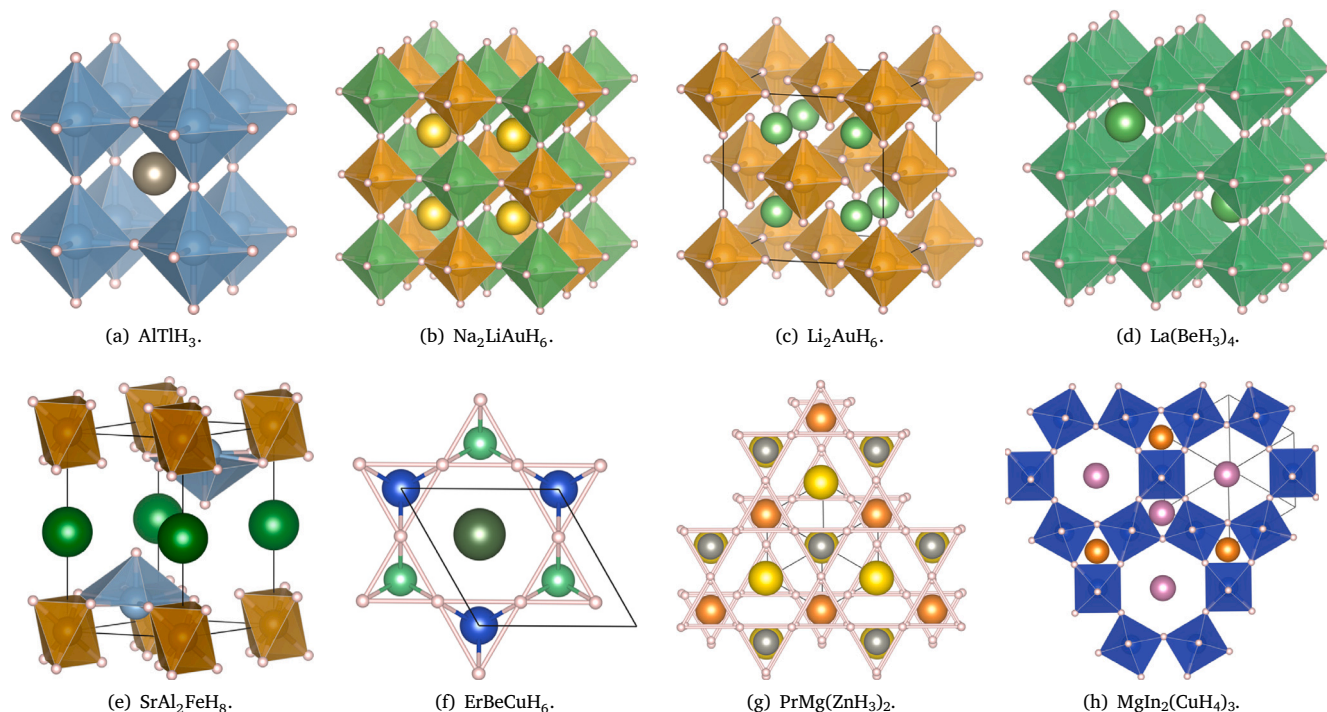


Fig. 5. Crystal structures representing some of the most represented structures. The unit cell is represented in black lines. In all the images H corresponds to pink, Ti and Zn gray, Al light blue, Na yellow, Li lime, Fe and Au orange, Be green, Cu blue, Mg orange, In pink, Pr yellow and all the other lanthanides as green.

values of both phases are comparable, with 46 K for the perovskite and 42 K for the $P3m1$ structure. However, the perovskite phase shows a significantly higher $\omega_{\log}$ of 627 K versus 436 K in the alternative structure. This reduction is offset by an increased electron–phonon coupling strength of $\lambda = 1.186$ in the trigonal phase, compared to $\lambda = 0.987$ for the perovskite.

3.3.3. Vacancy-ordered double perovskites

Hydride vacancy-ordered double perovskites, with the general formula SM_2-TM-H_6 (e.g., Li_2AuH_6 , SI #199), where SM is typically from groups IA–IIIA, have previously been reported in Refs. [26,39,40,64]. Similar to the double perovskites, these compounds crystallize in a highly symmetric face-centered cubic structure (space group $Fm\bar{3}m$), with SM atoms in the B sites, and TM in half the A sites. In this structure, hydrogen atoms form isolated H_6 undistorted octahedra surrounding each TM (see Fig. 5(c)).

The electronic structure of these compounds is characterized by a few bands crossing the Fermi level, often dominated by hydrogen and TM states. This is particularly evident when the simple metal belongs to the groups IA and IIA. In what concerns the phonons, the low-frequency spectra are composed of modes from the heavy SM and TM atoms, sometimes hybridized with hydrogen modes. The high-frequency modes, fully separated from the lower manifold, originate from localized hydrogen vibrations within the H_6 octahedra and dominate the contribution to the Eliashberg spectral function.

In addition to the compounds previously reported, such as Mg_2PtH_6 (SI #191), Mg_2IrH_6 (SI #185), and Li_2CuH_6 (SI #198) [26,39,40], we also identified new high- T_c candidates within this family. One example is Na_2BeH_6 (SI #201), with a predicted T_c of 51.6 K. This compound is high above the convex hull ($E_{\text{hull}} = 373$ meV/atom), and has a very large coupling constant ($\lambda = 1.921$). The electronic band structure of Na_2BeH_6 is dominated by hydrogen states near the Fermi level, and is very similar to the double perovskite Na_2LiCuH_6 . In fact, it also presents two bands crossing along the $\Gamma-L$ line. The phonon spectrum is separated into two main regions. The low-frequency modes are

composed of vibrations from Na, Be, and some hydrogen, all of which contributing significantly to λ . These are followed by high-frequency modes from hydrogen.

A different type of vacancy-ordered perovskites where the simple metal sites are occupied by two distinct species (see Fig. 5(d)), with the general form $SM^1-SM^2-TM-H_6$, is also present in our dataset. The highest T_c corresponds to $MgAlFeH_6$ (SI #395) with a T_c of 43.8 K. The electronic band structure presents a single band crossing the Fermi level, with two very evident flat bands between the L and U points, with DOS at the Fermi level being dominated by iron and aluminum states (see Fig. 6(a)). The phonon band structure is similar to previously reported systems of this family, with the lower-frequency modes corresponding to heavy atoms giving a modest contribution to λ .

We also identified a set of structures resembling perovskites but with half of the Wyckoff 1a sites vacant. Most of these compounds have the general chemical formula $LN(SMH_3)_4$ (e.g., $La(BeH_3)_4$, SI #35), where SM is either beryllium or zinc, and LN a lanthanide or Sr. The density of states at the Fermi level of the beryllium-containing compounds is primarily dominated by hydrogen states. In many cases, we observe a band gap between the conduction bands, indicating that these compounds can be interpreted as degenerate hole-doped semiconductors. The phonon band structure exhibits two separate manifolds of bands. The lower manifold is composed of (in order of increasing frequency) the LN atom, then Be, and finally Be–H hybridized vibrations, these latter being the main contributors for λ . The upper branches correspond to high-energy vibrational modes of the hydrogen atoms. The highest T_c for this group corresponds to $Sr(BeH_3)_4$ (SI #34) with a value of 76.7 K.

3.3.4. Isolated octahedra

A frequently occurring composition in our dataset is $LN-M_2-TM-H_8$ (e.g., $SrAl_2FeH_8$, SI #106), where M is a metal, TM corresponds predominantly to late transition metals, most often Fe, Cu, and Pd, and LN denotes lanthanides and sometimes elements

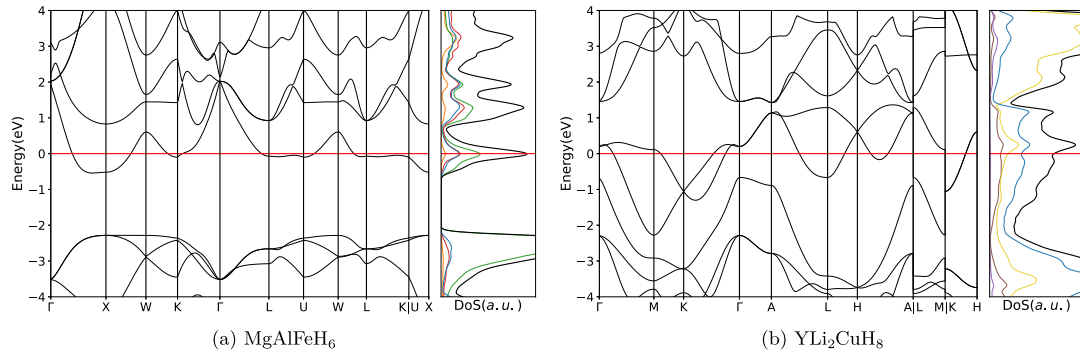


Fig. 6. Electronic band structures and density of states for two systems that exhibit flat bands (a) MgAlFeH_6 and (b) YLi_2CuH_8 . The black lines denote the bands in the bands structures and the total density of states, while the projected atomic density of states for hydrogen is represented by blue, iron by green, magnesium orange, aluminum red, lithium purple, copper brown and yttrium by yellow.

from groups IIA and IIIB of the periodic table. These compounds crystallize in layered structures consisting of alternating $\text{M}_2\text{-TM-H}_8$ layers separated by LN atoms. The M-TM-H_8 layers are composed of interconnected polyhedra centered at the M and TM atoms. Hydrogen atoms surround the TM atoms, forming octahedra that are linked through polyhedra centered at the M atoms (see Fig. 5(e)). The specific type of polyhedron formed around M depends on its chemical identity. When M is a simple metal from groups IA or IIIA, the TMH_6 octahedra are arranged to form a hexagonal pyramid around M, sharing common edges. In contrast, when M is a transition metal, the resulting polyhedron around the smaller M adopts a trigonal pyramidal geometry, connected to the TMH_6 octahedra through shared vertex.

Despite the chemical diversity across different choices of LN and TM, the electronic structure remains largely governed by the metal M. For a fixed M, compounds often share very similar band structures. In particular, most of the Ga-based compounds (e.g., $\text{SmGa}_2\text{FeH}_8$, SI #121) exhibit three bands crossing the Fermi surface, two of which crossing at the H-point. This compound can be understood as a degenerate electron-doped compound, as it exhibits a gap close to the top of the valence. In addition, a relatively flat band can be seen in the vicinity of the Fermi level between Γ and M (see Fig. 6(b)), especially in lithium-based systems (e.g., YLi_2CuH_8 , SI #132).

The phonon spectra are composed of three separate manifolds: the lowest-energy modes are due to the heavy atoms (LN, M, and TM), while hydrogen vibrations occupy the middle and high-frequency ranges. In gallium-based systems, the hydrogen modes are well separated from the rest, whereas in compounds based on lithium, partial overlap between hydrogen and lithium vibrations is observed. There are comparable contributions to λ from both hydrogen and the heavier elements.

Overall, compounds in the $\text{LN-M}_2\text{-TM-H}_8$ family exhibit superconducting critical temperatures T_c ranging from 27 K to 46 K.

3.3.5. Kagomes

Kagome materials have been a focus of research in high-temperature superconductivity due to their geometric and topological properties [66,67]. These intrinsic properties have also been examined in hydride systems [68], and some high-pressure hydride kagome phases have been reported [69]. Wang et al. proposed ThSiH_7 as a dynamically stable kagome at ambient pressure with $T_c = 61.5$ K [70]. Recently, Ishikawa et al. explored lanthanum hydrides and identified one dynamically stable system at ambient pressure with $T_c = 18.67$ K [71].

Several systems featuring hydrogen in a breathing kagome lattice appear in our dataset, with the most common having the ternary composition LN-SM-TM-H_6 (e.g., ErBeCuH_6 , SI #468). In this formula, LN typically represents lanthanides, though some elements from groups IA to IIIB can also occupy this position, while TM corresponds to late

transition metals such as Cu or Pd. These compounds predominantly crystallize in the $\bar{P}6m2$ space group, where the structure features chains of trigonal prisms sharing hydrogen corners, forming the characteristic “star of David” pattern of kagomes (see Fig. 5(f)). One additional phase occurs with the $P3m1$ space group (e.g., NaCaCuH_6 , SI #484), where two distinct breathing kagome layers are shifted relative to each other.

The kagome compounds with the highest T_c have general formula LiLNpDH_6 , which interestingly, may crystallize in either the $\bar{P}6m2$ phase or the $P3m1$ phase (e.g., LiDyPdH_6 , SI #491). The $\bar{P}6m2$ phase is generally associated with higher T_c values and larger E_{hull} , while the $P3m1$ structure tends to be thermodynamically more favorable, though with reduced T_c .

The electronic structure of the $\bar{P}6m2$ compounds typically includes two bands crossing the Fermi surface. The Fermi level sits between two peaks in the density of states, mostly derived from lanthanide and hydrogen states. The phonon spectrum is divided into low-frequency modes (from Pd and Ln), a very localized mid-frequency region (composed of Li vibrations) around 55 meV that partially overlaps the high-frequency region above (from H states), with substantial contributions to λ from both low energy and high energy vibration modes. The $P3m1$ band structure is similar to the $\bar{P}6m2$ phase, also exhibiting two bands intersecting the Fermi surface. However, the Fermi level now lies near a broad shoulder-like peak in the total DOS, resulting from hybridized states involving H, Pd, and Ln atoms. The phonon dispersion reveals partial overlap between Li- and H-modes in the mid to high-frequency range, reducing the effectiveness of vibrational decoupling seen in the higher-symmetry structure. These structural differences lead to distinct superconducting performance: the $\bar{P}6m2$ phase typically achieves T_c values around 50 K, while $P3m1$ compounds are closer to 30 K. This difference stems primarily from a reduced electron-phonon coupling in the latter, as both structures exhibit similar ω_{log} values.

Another well represented group of kagomes with the composition $\text{SM-LN(TM-H}_3)_2$ is also recurring in our dataset (e.g., $\text{PrMg(ZnH}_3)_2$, SI #284). These materials exhibit a structure with space group $R\bar{3}m$, crystallizing in layers of pairs of breathing kagome patterns that are shifted from each other (see Fig. 5(g)). The electronic and vibration properties of this group are dependent on the TM atom in the center of the triangles, being the highest T_c associated with gold (only occurring with lithium as the SM atom). Those compounds present an electronic band structure with one or two bands crossing the Fermi surface, overlapping in some points. The phonon band structure presents some well separated manifolds of different vibrational modes. At the lower frequencies are the gold and lanthanide modes, followed by two separated groups of bands corresponding to lithium, and then two manifolds coming from hydrogen vibrations. The metallic atoms contribute as much to λ as the hydrogen atoms, being the lithium vibration modes almost non-contributing.

Another family of kagome hydrides with general formula $\text{TM-SM}_2\text{-M}_3\text{-H}_{12}$ is also present in the dataset (e.g., $\text{MgIn}_2(\text{CuH}_4)_3$,

SI #16). These hydrides crystallize in the $R\bar{3}m$ space group, and display chains of hydrogen tilted octahedras that assemble into breathing kagome-patterned layers. These then alternate with layers exhibiting a rhombitrihexagonal tiling when seen from above the kagome pattern (see Fig. 5(h)). These compounds typically exhibit T_c of approximately 40–50 K when M is Zn and 20–30 K when M is Cu. The exception to this rule is $Mg_3S_2(CuH_4)$ (SI #26), which reaches the highest $T_c = 61.8$ K. The electronic and phonon band structures of this compound are complex, with multiple bands crossing the Fermi level and vibrational modes of light and heavy atoms overlapping. Nevertheless, the heavy atoms provide the dominant contribution to the large electron–phonon coupling constant, $\lambda = 1.78$.

We also identified a regular kagome phase, $Sc(CuH_3)_2$ (SI #203), with $T_c = 49.6$ K, and a breathing kagome variant $LiTmNiH_6$ (SI #396), with $T_c = 20.1$ K, which alternates the prism-centered atom between layers.

4. Discussion

We employed a machine learning accelerated workflow to predict the superconducting properties of over two million hydride structures, performing approximately 5900 density-functional perturbation theory calculations and identifying more than 600 dynamically stable materials with critical temperatures above 20 K. This exploration expands the known space of ambient-pressure hydride superconductors and reveals important insights into the mechanisms underlying high-temperature superconductivity in these systems.

Our dataset encompasses a large diversity of chemical compositions and structural motifs, including perovskites, double perovskites, vacancy-ordered perovskites, kagome lattices, compounds featuring isolated hydrogen-octahedra, etc. Despite this structural and compositional variety, compounds with high critical temperatures consistently exhibit a significant contribution of hydrogen states to the density of states at the Fermi level. The correlation between T_c and hydrogen-projected DOS normalized by volume highlights that achieving high superconducting temperatures requires not merely the presence of hydrogen, but its active involvement in the electronic states responsible for Cooper pairing.

An interesting characteristic of many high- T_c candidates in our dataset is their resemblance to degenerate semiconductors. Numerous compounds exhibit band structures with band gaps near the Fermi level, effectively behaving as heavily hole- or electron-doped systems. This observation confirms that the superconducting state emerges from a delicate balance between localized bonding states and itinerant metallic behavior, reminiscent of doped semiconductors. The prevalence of this electronic motif across diverse structural families indicates that chemical or physical doping strategies could serve as powerful tools for engineering superconductivity in hydride materials.

While the mechanism of superconductivity in covalent-bond-driven metals has been well understood since the discovery of MgB_2 [72], what emerges from our extensive search is that this mechanism appears as a universal route to high critical temperatures across systems that share no structural similarity with MgB_2 . The convergence of our unbiased search on this electronic signature suggests that covalent bonding may represent the only pathway to achieving high- T_c conventional superconductivity.

None of the identified superconductors lies on the convex hull of thermodynamic stability at ambient pressure. The distribution of distance to the hull, centered around 200 meV/atom, indicates that these phases face substantial barriers to synthesis under equilibrium conditions. This represents the primary obstacle to practical realization of ambient-pressure hydride superconductors. The scarcity of thermodynamically stable high- T_c hydrides at ambient pressure reflects well-established chemistry. Hydrogen-rich phases are generally unstable with respect to H_2 release at ambient pressure due to the extremely strong H–H bond (436 kJ/mol) and high entropy of the hydrogen

gas. This is why nearly all experimentally known high- T_c hydride superconductors, including H_3S , LaH_{10} , YH_9 , and CaH_6 , require high pressure (≥ 100 GPa) for stabilization and decompose upon decompression. Our computational survey confirms and emphasizes this trend: the structural motifs that favor high T_c (dense hydrogen sublattices, specific metal-hydrogen coordination) are precisely those that are thermodynamically disfavored at ambient pressure. Nevertheless, our findings provide a starting point for potential stabilization strategies. The observation that many compounds resemble doped semiconductors suggests that introducing doping controlled in thermodynamically stable semiconducting hydrides, either through chemical substitution or electrostatic gating, could provide a viable synthesis route. Additionally, the application of moderate pressures or elevated temperatures during synthesis may stabilize metastable superconducting phases, which could then be retained upon quenching to ambient conditions, analogous to the synthesis strategies employed for metastable materials like diamond or certain perovskites.

At this point, we emphasize the critical importance of computing thermodynamic stability using the most complete convex hull possible. Incomplete phase diagrams artificially reduce formation energies above the hull, leading to overoptimistic assessments of synthesizability that can mislead experimental efforts. Two recent examples illustrate this issue.

Huang et al. [73] studied hydride compounds based on the fluorite-type XH_8 structure, reporting $AcRhH_8$ and $BaRhH_8$ as stable under ambient pressure with T_c values of 78 K and 52 K, respectively. Comparison with our dataset reveals that while $BaRhH_8$ (SI #168) exhibits a high T_c of 49.6 K, the material is thermodynamically unstable, with $E_{hull} = 104$ meV/atom. The compound $AcRhH_8$ is even more unstable, with $E_{hull} = 206$ meV/atom. These discrepancies suggest that the phase diagrams reported by the authors are highly incomplete.

Quan et al. [74] investigated perovskite hydrides with the general formula $MScH_3$ ($M = K, Rb, Cs$), proposing the possibility of synthesizing such metastable phases via ball milling, analogous to the method used for $KMgH_3$. This approach was justified by the highly negative formation energies calculated in their work. Our dataset, however, presents a contrasting scenario. While $KMgH_3$ exhibits an energy above the convex hull of only 5 meV/atom, the scandium-containing analogs $KScH_3$, $RbScH_3$, and $CsScH_3$ display substantially higher energies of 223 meV/atom, 253 meV/atom, and 309 meV/atom, respectively, which significantly diminish the likelihood of successful synthesis.

The most comprehensive convex hulls currently available for thermodynamic stability at ambient pressure are those from GNoME [75] and Alexandria [26–28]. Both databases are freely accessible (although GNoME has a non-commercial license), and provide millions of calculated structures spanning diverse chemical spaces. Given their availability, researchers have no justification for relying on incomplete or *ad hoc* reference datasets when assessing material stability. These comprehensive databases offer a consistent and reliable baseline that enables accurate predictions of thermodynamic stability and synthesizability, thereby preventing the overoptimistic assessments that can arise from incomplete phase diagrams. We strongly recommend that future computational studies of novel materials use these resources to ensure their stability predictions reflect the full complexity of competing phases.

The detailed structural and electronic analysis presented here identifies specific chemical and structural features that promote high critical temperatures, providing experimentalists with concrete targets for synthesis efforts. The recurring motifs, such as hydrogen kagome lattices, isolated H_6 octahedra, and specific coordination geometries, offer templates for rational materials design. The strong representation of late transition metals such as Cu, Pd, Pt, and Au and simple metals such as Li, Na, Al, and Ga in our dataset highlights promising chemical spaces for future exploration.

While our study provides an extensive exploration of the hydride superconductor landscape, several important limitations must be acknowledged. First, despite screening over two million structures, our

dataset remains inherently incomplete. The chemical and structural space of possible hydrides is combinatorially vast, and many potentially promising compositions and crystal structures, particularly those with large unit cells or complex stoichiometries, remain unexplored. Second, our theoretical framework, based on density functional theory with a semi-local functional and Migdal–Eliashberg theory, has well-known limitations. The semi-local functional may not accurately capture electronic correlations in certain transition metal hydrides, particularly those containing localized d or f electrons. Furthermore, the Migdal–Eliashberg formalism assumes weak to moderate electron–phonon coupling and may break down for systems with very large λ , where vertex corrections and non-adiabatic effects beyond the Born–Oppenheimer approximation become significant. Computing the full solution of the anisotropic Eliashberg equations would result in more accurate solutions, however, solving such equations for ~ 5900 materials is computationally prohibitive. Recent benchmarks on binary hydrides suggest that the Allen–Dynes formula typically underestimates T_c by ~ 5 – 10 K relative to Eliashberg solutions for $\lambda < 1.5$ but can deviate further for $\lambda > 2$ [40]. Since approximately 80% of our dataset has $\lambda < 1.5$, we expect the Allen–Dynes approximation to be reasonable for the majority of compounds and for the purpose of large scale screening. Our approach also cannot account for unconventional pairing mechanisms beyond the conventional BCS framework, which could be relevant in systems with strong electronic correlations or exotic band structures. Third, we treat all materials within the harmonic approximation, neglecting anharmonic phonon effects that can be substantial in hydrogen-rich compounds. Such anharmonicity could significantly alter both the phonon spectra and electron–phonon coupling strengths and therefore T_c . Fourth, our calculations assume perfect stoichiometry and crystalline order, while real materials inevitably contain defects, vacancies, and compositional variations. Moreover, we do not explore the effects of chemical doping, which, as our electronic structure analysis suggests, could be a crucial pathway to stabilizing these phases and tuning their superconducting properties. Controlled introduction of charge carriers through substitutional doping or intercalation might not only enhance thermodynamic stability but also modify the electronic structure in ways that could raise T_c . Finally, we focus exclusively on equilibrium structures at zero temperature and ambient pressure, neglecting the possibility that some compounds might be stabilized through kinetic trapping, epitaxial strain in thin films, or moderate pressure/temperature conditions. Despite these limitations, our high-throughput approach successfully identifies systematic trends and design principles that can guide both more sophisticated theoretical treatments and experimental synthesis efforts toward realizing ambient-pressure hydride superconductors.

In summary, while our comprehensive search reveals that achieving both high critical temperatures and thermodynamic stability at ambient pressure remains elusive, the extensive dataset and design principles establish a foundation for future experimental and theoretical efforts. The path forward lies not in abandoning the search for ambient-pressure hydride superconductors, but in developing innovative synthesis strategies, including doping, high-pressure or high-temperature synthesis followed by quenching, and thin-film deposition techniques, that can overcome thermodynamic barriers while preserving the favorable electronic structures identified in this work. Our findings demonstrate that the theoretical landscape of high- T_c hydride superconductors is vast and rich, but has to wait for experimental realization through creative synthesis approaches.

CRediT authorship contribution statement

Paulo R. Pires: Writing – original draft, Visualization, Investigation, Formal analysis. **Thalis H.B. da Silva:** Writing – original draft, Investigation, Formal analysis. **Kun Gao:** Writing – original draft, Investigation. **Kaja H. Hiorth:** Writing – original draft, Investigation. **Tiago F.T. Cerqueira:** Writing – original draft, Investigation. **Théo**

Cavignac: Writing – original draft, Visualization, Investigation. **Pierre-Paul De Breuck:** Writing – original draft, Investigation. **Hai-Chen Wang:** Writing – original draft, Visualization, Investigation. **Đorđe Dangić:** Writing – original draft, Investigation. **Yue-Wen Fang:** Writing – original draft, Investigation. **Antonio Sanna:** Writing – original draft, Investigation. **Wenwen Cui:** Writing – original draft, Investigation. **Ion Errea:** Supervision, Investigation, Funding acquisition. **Päivi Törmä:** Supervision, Investigation, Funding acquisition. **Miguel A.L. Marques:** Supervision, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Research Link Provided

[Geometries optimized with PBEsol \(Original data\)](#) (Alexandria)

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